INF 345 — Fundamentals of DevOps

Lecture 1: Welcome & The DevOps Landscape

Adil Akhmetov · Lesson 1

Today’s agenda

- ▢ What DevOps actually means (not just "the tools")
- ▢ The toolchain map for this entire course
- ▢ How the Red Hat Academy partnership fits in
- ▢ The syllabus — grading & weekly plan
- ▢ How grading works (autograded, fast feedback, no surprise waits)
- ▢ What to do before next lecture

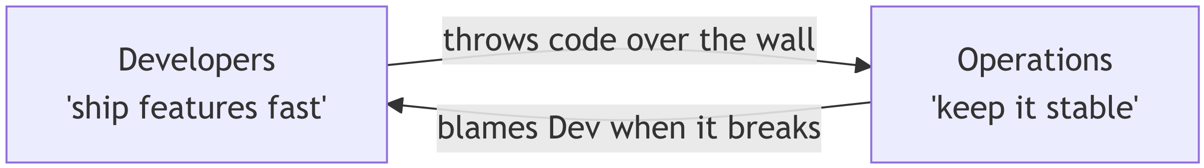
"But it works on my machine."

If you've ever heard this — or said it — you already understand the problem this entire course exists to solve.

Block 1

What is DevOps, really?

The wall between Dev and Ops



Two teams, opposite incentives, one production system stuck in the middle. DevOps exists because this wall was a organizational failure, not a technical one.

DevOps is a culture, not a tool — CALMS

- **C**ulture — shared ownership, no blame-throwing over the wall
- **A**utomation — humans shouldn’t repeat what a script can do reliably
- **L**ean — small batches, fast feedback, cut waste
- **M**easurement — you can’t improve what you don’t measure
- **S**haring — knowledge and tooling belong to the whole team, not one hero

Every tool in this course (Podman, Ansible, GitHub Actions) is an implementation detail of one or more of these letters — not the point itself.

Quick discussion

Where has "it works on my machine" bitten *you*,
or someone you know?

(No wrong answers — this is the pain the rest of the semester fixes.)

Block 2

This course's toolchain

The map for this semester

Each box is one module of this course. Nothing here is optional trivia — each module's lab builds directly on the last one.

Why these specific tools

Podman — CLI-compatible with Docker

ANSIBLE

Ansible — already fully open-source

GitHub Actions — free tier, industry-standard

Rule of thumb for this whole course: whenever a lab uses a Red Hat–specific tool, the README also tells you the free/open equivalent. Nothing here is locked to one vendor.

Where Red Hat Academy fits in

- SDU partners with Red Hat Academy (RHA) — this gives you **free** access to official Red Hat cloud labs for two modules: Containers (Podman) and Automation (Ansible).
- **Your grade for those two labs comes from finishing the RHA lab itself** — your instructor assigns it from the RHA portal, not from a GitHub check.
- These are the same labs used in paid corporate Red Hat training — normally not free.
- A few students each year get a shot at a real Red Hat certification voucher through this partnership.
- CI/CD isn’t covered by RHA at all — that module runs entirely on GitHub, which you’d need anyway for literally any software job.

Block 3

How this course runs

The syllabus, in one slide

Course: INF 345 — Fundamentals of DevOps **Credits:** 3 cr / 5 ECTS (2 lecture + 1 seminar + 2 lab hrs/week) **Mode:** Online · **Type:** Elective **Format:** 15 lessons, lecture + 1h practice each

Final Exam: delivered as a Capstone Project + demo (policy allows a project to substitute for a written exam) **Full doc:** `SYLLABUS.md` in the repo root **Questions:** office hours, or email (see syllabus for etiquette)

Read the full syllabus before Lesson 2 — it has the complete weekly plan, assessment rubric pointers, and academic integrity policy.

Weekly plan — Fall 2026

Wk	Topic
1	Welcome & the DevOps landscape
2	Git/GitHub workflow & Linux CLI
3	Containers 101: why & how
4	Images, layers, multi-stage builds
5	Container networking & volumes
6	Container security — **Lab 01 due**
7	Config management & Ansible basics
8	Playbooks, roles, variables

Wk	Topic
9	Idempotence & testing (Molecule)
10	Ansible in production — **Lab 02 due**
11	CI/CD concepts & GitHub Actions
12	Building & testing pipelines
13	Security scanning — **Lab 03 due**
14	Capstone integration work session
15	Capstone demos — **Final Exam due**

Full detail, rubrics, and policies: `SYLLABUS.md` in the repo.

Grading breakdown

Every lecture has the same shape

1. **Recap** — 2-3 quick questions from last time (spaced repetition)
2. **Agenda** — today's roadmap as a checklist, always visible
3. **Hook** — why today's topic actually matters, before any theory
4. **Chunked blocks** — with a real break in between, ending early if we're done
5. **"Before next lecture"** — a concrete, short prep checklist

This isn't arbitrary — short chunks, fast feedback, and a predictable structure just work better for everyone.

Fast feedback, not a week of waiting

- Every lab lives in your own GitHub Classroom repo.
- Push your work → GitHub Actions runs the autograder → you get a pass/fail/partial score **within minutes**.
- No submitting into a void and waiting for grades next week.

Before next lecture

Lesson 2 is Git & GitHub — come with these ready:

- ☐ Make sure you have a GitHub account
- ☐ Accept the GitHub Classroom invite (link posted separately)
- ☐ Install Docker Desktop **or** Podman Desktop on your laptop
- ☐ Read `SYLLABUS.md` in full

~15-20 min of prep, not a lab.

See you next lecture

Git & GitHub — the version control workflow every lab in this course runs on.